CSIS 2430 —Project Plan (Plan + Trace)

1) Project Information

- Course / Term: CSIS 2430
- Project: Project 3
- Team Name: Team 3
- Start Date: 2026-03-23
- Due Date: 2026-04-10
- Checkpoint Date (mid-project): 2026-04-03

2) Team & Roles (assigned for this project)

Roles are assigned per project and rotate between projects.

Team Member	Role (Implementation / Verification / Communications)	Primary Responsibilities (1–2 bullets)	Availability / Notes
Kevin	Implementation Lead	• Brute Force / Exhaustive Search; • Comparison Summary Output.	Friday Evening or Weekends
Latifah	Co-Implementation Lead	• Implement Three Greedy Strategies; • Comparison Summary Output.	Friday Evening or Weekends
Jimmy	Verification Lead	• Test plan, Edge cases, Verification evidence.	Friday Evening or Weekends
Agnes	Communications Lead	• Report assembly, • Deliverable packaging.	Friday Evening or Weekends

3) Repository & Access (Required for Projects 1–4)

- Platform: GitHub
- Repository URL: https://github.com/agnesbfr/CS2430-Project3.git
- Instructor has access to repository? Yes. Repository is Public.

- Repo Structure Requirement (summary): Yes. contributions.md; project_plan.md, uml.png, pseudocode.md. Available at: https://github.com/agnesbfr/CS2430-Project3/tree/main/docs

4) Milestones (Timeline)

Milestone	Target Date	Owner	Definition of Done	Status (NS/IP/D)	Evidence Link / Pointer
Repo set up	2026-03-23	Agnes	Repo created; README run steps; sample I/O runs.	D	CS2430-Project3/docs/
Core implementation complete	2026-04-01	Kevin/Latifah	All required features implemented; compiles/runs end-to-end.	D	CS2430-Project3
Verification complete	2026-04-08	Jimmy	Test suite covers edge cases; expected vs actual documented.	D	CS2430-Project3
Report/video draft complete	2026-04-08	Agnes	Draft report sections complete; video outline + script notes.	NS	
Final integration + submission	2026-04-09	Agnes	All deliverables finalized; tags release; submission package ready.	NS	

5) Task Backlog (Assignments + Status)

ID	Task	Owner	Due Date	Status (NS/IP/D)	Evidence Link / Pointer	Notes / Dependencies
1	Create repository + baseline README/run steps	Agnes	2026-03-23	D	CS2430-Project3	None
2	Draft design artifacts	Kevin/Latifah	2026-03-26	D	CS2430-Project3/docs/	None

3	Track team task completion and status updates	Agnes	2026-04-06	D	docs/project_plan.md	Being updated by the team when tasks are done.
4	Implement core data structures and parsing	Kevin/Latifah	2026-04-01	D	CS2430-Project3/Project3	Review phase
5	Implement edge-case tests + expected outputs	Jimmy	2026-04-01	D	CS2430-Project3	Waiting implementation part be done
6	Checkpoint package: update plan + docs + invite verification	Agnes	2026-04-03	D	CS2430-Project3/docs	Checkpoint submission
7	Code Quality Check	Jimmy	2026-04-07	NS		Responsible to review the codings
8	Report complete: GitHub, video, packaging	Agnes	2026-04-09	NS		

6) Verification Plan

- Correctness tests: Validate greedy strategies and brute-force results; compare outputs to ensure correctness.
- Edge cases: Empty input, items exceeding weight limit, multiple optimal solutions, and total weight = 700 kg.
- Expected vs. actual: Document sample outputs and compare greedy results with the optimal solution.
- Performance: Ensure brute force evaluates all subsets and compare runtime with greedy strategies.

7) Risks / Blockers

Risk / Blocker	Impact	Mitigation Plan	Owner	Status
Coding part be incorrect.	High	Kevin and Latifah are working hard on the coding part, and also we will	Jimmy	Active

		have Jimmy reviewing everything before the project due date.		
Incomplete or unclear documentation before submission.	High	Communications Lead reviews report, and /docs artifacts before deadline.	Agnes	Active

8) Update Log

Date	Summary of Update (what changed / what completed)	Blockers / Resolutions	Link(s)
2026-04-01	Repository created, README added, and initial skeleton program executed successfully.	None.	CS2430-Project3
2026-03-26	Designed artifacts in the repository: project_plan.md, uml.png, pseudocode.md.	Updated.	CS2430-Project3 /docs